Causal DAG-Building Roadmap — Final Methodology (Pre-Implementation)

Status: nothing in this document has been implemented. This is the design for the next phase of the project, to be executed once agreed. Every "what we do" step below is a planned action, not a completed one.

This document supersedes the earlier DAG-building sketch. The critique was substantively correct on several points — most importantly that causal discovery algorithms were at risk of being asked to answer data-semantics questions they cannot answer, that algorithmic output was at risk of being treated as more authoritative than it is, and that `z_ecg`'s causal role needed to stay genuinely open rather than resolved by a hard-coded preprocessing choice. The restructuring below reflects that. A few places where the critique's suggestions needed further sharpening (not just acceptance) are noted explicitly in Phase-level "what must NOT be assumed" sections and in the Final Methodological Judgment.

Phase 0 — Causal Estimand and Problem Definition

Objective: Fix what question the DAG is being built to answer. A DAG is not a property of a dataset in isolation — it is a property of a dataset *given a specific treatment, outcome, and target population*. Without this fixed first, "build the DAG" is not yet a well-posed task.

Inputs: Your earlier candidate causal questions (treatment/outcome/population triples), the master dataset's variable list.

What we do:

- Select (or carry forward as still-open) the finalist candidate causal question(s) from the earlier planning stage — specifically: candidate treatment variable(s), placeholder outcome definition, target population, and estimand (ATE vs. CATE as the eventual goal, noted now so Phase 11's adjustment-set derivation is built for the right target).
- Explicitly write down what is *not yet decided* about the estimand (e.g., if outcome-label provenance is still unresolved, that unresolved state is carried forward as a tracked dependency, not silently assumed away).

Decision/output produced: A short, explicit statement: "the DAG we are about to build targets treatment T, outcome Y (placeholder/finalized), population P." If multiple finalist candidates remain live, one DAG-building pass per candidate is required — they are not interchangeable, and a single "general" DAG across all candidates would hide treatment-specific confounding structure.

What must NOT be assumed: That a single DAG serves every candidate causal question equally well. That the outcome definition is settled if it isn't — carry the placeholder forward rather than quietly treating `is_MI` as final.

Validation/checkpoint: Confirm with yourself (and ideally a quick check with the expert, even informally, before the heavier phases) that the candidate question(s) chosen here are still the ones worth investing discovery/literature/expert-validation effort into.

Phase 1 — Causal Unit and Temporal Framework

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Objective: Fix the unit of analysis and the temporal skeleton the DAG will be built on top of. This is a data semantics decision and must precede any discovery algorithm, per the critique — correctly so, because PC/GES/FCI have no mechanism for inferring "what counts as one observation" or "what counts as before vs. after."

Inputs: Master dataset's ECG-per-patient/admission structure, `ecg_time`, `index_mi_time`, `time_diff_hours`.

What we do:

- Decide patient-level vs. admission-level as the causal unit, based on: how many admissions per patient exist, whether treatment/outcome timing is more naturally admission-scoped or patient-scoped, and whether the split already performed (`SubjectID`-level) remains valid under either choice (it does, since admission-level units nested in patient-level split groups are still split-safe).
- Define **time zero** precisely — the event that separates "baseline" from "everything downstream." Candidate: index-ECG time. State explicitly whether time zero is ECG-time or a separate treatment-assignment time, since these can differ.
- Define the **index-ECG selection rule** for patients/admissions with multiple ECGs: which ECG is "the" baseline ECG, and what happens to the others (discarded? used as a secondary/robustness covariate? used in a different, explicitly out-of-scope time-varying analysis?).
- Run the **temporal sanity checks**: sign convention and distribution of `time_diff_hours`, whether any index-ECG candidates fall at/after `index_mi_time`, counts of ECGs per unit under the candidate selection rule.

Decision/output produced: A written, versioned specification: causal unit = {patient | admission}; time zero = {definition}; index-ECG rule = {definition}; a data table showing how many units survive under this rule and how many ECGs/units were excluded and why.

What must NOT be assumed: That "before"/"during"/"after" in any variable name reflects the time-zero you just defined — that mapping is Phase 2's job, not this phase's. That multiple ECGs per unit can be silently averaged or the "first" one chosen without checking whether "first" is clinically meaningful (e.g., first ECG in the encounter could predate symptom onset and be uninformative).

Validation/checkpoint: Spot-check a sample of units by hand against the chosen rule before proceeding — this is cheap and catches rule-definition errors that would otherwise propagate through every later phase.

Phase 2 — Treatment and Outcome Audit

Objective: Determine, with evidence rather than naming convention, which treatment variables are genuinely eligible as *baseline* treatments relative to the Phase 1 time zero, and resolve (or explicitly flag as unresolved) the outcome definition.

Inputs: Phase 1 output, `_before` / `_during` / `_after` treatment columns, `Label_Troponin`, `is_MI`, `index_mi_time`.

What we do:

- For each `_before` treatment candidate: look for any available timestamp evidence (even indirect, e.g., linkage to MIMIC-IV prescription tables) confirming the flag reflects status strictly before time zero for every row, not just "typically." Where no such evidence exists, document the flag as an **unverified**

temporal assumption, not a confirmed one.

- For `_during` / `_after` variables: confirm (do not assume) what encounter window "during" spans relative to time zero, and whether any of them could in principle still be used as a treatment under a *different*, explicitly reframed causal question (e.g., a question where time zero is later in the encounter) — but do not fold that reframing into the current Phase 0 estimand without a deliberate decision to do so.
- Resolve, or document as still-open, the outcome-label provenance question (`Label_Troponin` vs `is_MI` vs any further "dataset ground truth" label) — including whether `index_mi_time` / `time_diff_hours` are themselves derived partly from the outcome label (a leakage risk if so).

Decision/output produced: A table: treatment candidate → {temporally verified / verified-via-indirect-evidence / unverified-assumption} → eligibility verdict for baseline-treatment use. A written outcome-definition status (resolved, with derivation documented, or explicitly open with the specific open question stated).

What must NOT be assumed: That suffix naming (`_before`) is itself evidence. That an unresolved outcome definition can be worked around by proceeding with a placeholder indefinitely — this phase should produce either a resolution or an explicit blocking dependency for Phase 0.

Validation/checkpoint: No treatment variable proceeds to Phase 3 eligibility without an explicit verification tag. No DAG-building work proceeds past this phase with a fully unresolved outcome definition (a *placeholder* outcome, clearly labeled as such, can proceed — a silently assumed one cannot).

Phase 3 — Baseline Covariate Eligibility

Objective: Determine which non-treatment variables are legitimately available *before* time zero, separately from deciding their causal role (which comes later, in discovery/literature/expert phases).

Inputs: Phase 1/2 outputs, comorbidity flags (`diabetes`, `hypertension`, `ckd`, `prior_mi`), demographics, ECG morphology features, `z_ecg`.

What we do:

- Check whether comorbidity flags reflect "known as of time zero" or "ever recorded in the full record" (the latter would include post-time-zero diagnoses — a leakage-relevant eligibility failure, not merely a labeling nuance).
- Confirm demographics are time-invariant or, if age is encounter-specific, that it's computed as of time zero.
- Confirm ECG morphology features and `z_ecg` are computed from the Phase 1 index ECG specifically (not a different/later ECG) for every unit.

Decision/output produced: An eligibility table: variable → {eligible as baseline / eligible with caveats / ineligible (post-time-zero)}. This table does **not** yet assign causal roles (confounder/mediator/etc.) — it only answers "is this variable even temporally available for baseline use."

What must NOT be assumed: That a variable being "baseline-eligible" means it's a confounder — eligibility is a necessary, not sufficient, condition for inclusion in the adjustment set. That `prior_mi` is free of ambiguity — confirm it specifically means *prior to this encounter/time zero*, since in some EHR extracts "prior" fields are computed relative to extraction time, not per-encounter.

Validation/checkpoint: Every variable destined for Phase 4's discovery dataset has passed this eligibility check first.

Phase 4 — Discovery-Ready Dataset Construction

Objective: Assemble the exact dataset the discovery algorithms (Phase 7) will run on, using only Phase 0–3 decisions — this phase is mechanical given prior phases, not a place for new judgment calls.

Inputs: Phase 0–3 outputs.

What we do:

- One row per causal unit (Phase 1).
- Columns: treatment (Phase 2-eligible), outcome (Phase 2 resolution/placeholder), baseline-eligible covariates (Phase 3), `z_ecg` representation (format depends on Phase 5 — see below, this may be provisional pending that phase).
- Exclude identifiers and raw time anchors as graph nodes (they define structure, not causal roles).
- Document final row count and any attrition from Phase 1–3 exclusions.

Decision/output produced: The discovery dataset itself, versioned, plus a data dictionary mapping every column to its Phase 1–3 justification.

What must NOT be assumed: That this dataset is final for *all* downstream purposes — the representation used here may differ from what's used in the eventual propensity/balance model (Phase 5 addresses this explicitly), so this dataset should be treated as discovery-specific, not silently reused unchanged for Phase 13.

Validation/checkpoint: Every column has a traceable justification back to Phases 1–3; no column fails eligibility but is included anyway "for now."

Phase 5 — `z_ecg` Representation and Causal-Role Analysis

Objective: This is treated as its own phase, not a preprocessing footnote, because the critique is right that the 64-dimensional embedding's causal role is a genuinely open, high-stakes question — and because the representation used for discovery, for the eventual propensity/balance model, and for prediction are allowed to differ and that distinction should be decided deliberately, not collapsed by default.

Inputs: `z_ecg_1...64`, ECG morphology features (for cross-referencing), Phase 1 timing (confirms whether `z_ecg` is computed from a pre-treatment ECG).

What we do:

1. **Confirm temporal eligibility first** (depends on Phase 1/3): is `z_ecg` computed from the index ECG, and is that ECG pre-treatment for every unit? If not uniformly true, `z_ecg`'s candidate role set shrinks immediately (a post-treatment embedding cannot be a baseline confounder, regardless of anything else).
2. **Separate the "for discovery" representation from the "for later use" representation as a deliberate decision, not a default.** Options, to be chosen based on empirical behavior rather than hard-coded in advance:

- Use all 64 dimensions directly in discovery if sample size and algorithm runtime tolerate it — test this empirically first rather than assuming it won't work.
- If instability or runtime is a real problem (determined empirically, e.g., by comparing discovery-graph stability across dimensionality choices), consider a reduction — but justify the specific reduction method (PCA, clustering, a representative/expert-nominated subset) by what it preserves for *this* purpose, and report how discovery-graph structure changes under more than one reduction choice as a robustness check, rather than committing to one method upfront.
- Whatever representation is used for discovery is documented as "discovery representation," explicitly allowed to differ from the representation later used in the propensity/balance model (Phase 13), which can be revisited once the DAG establishes `z_ecg`'s actual role.

3. **Pre-register the candidate causal roles** to be evaluated through Phases 7–10, rather than assuming one: baseline covariate, proxy for latent physiological state (candidate confounder), mediator (only plausible if timing in step 1 contradicts pre-treatment status), effect modifier, or predictive-only/no causal adjustment role.
4. **Explicitly test, where feasible, whether `z_ecg` (or its discovery representation) predicts treatment assignment** independent of the already-measured comorbidities/demographics. This is an empirical association check, not proof of a causal role — but a null result here is informative (it would weaken the confounder candidacy without ruling out proxy-for-severity/outcome-only roles), and a positive result is likewise only suggestive, not confirmatory, of a confounder role.
5. **Leakage check specific to `z_ecg`**: confirm the β-VAE was trained using only training-split, pre-time-zero ECGs. If this cannot be confirmed, document it as an unresolved leakage risk that must be flagged to the expert and in the final limitations, not silently set aside.

Decision/output produced: A `z_ecg` role-candidacy report: temporal eligibility verdict, discovery-representation choice (with justification and, ideally, a sensitivity comparison across ≥2 reduction choices if reduction was used), treatment-association test result, leakage-check result, and the pre-registered candidate-role list carried into Phases 7–10.

What must NOT be assumed: That `z_ecg` is a confounder because it predicts the outcome well (predictive strength is not evidence of a common-cause role — restated from the earlier roadmap and still the single most important caution for this variable block). That a fixed "reduce to 5–10 dimensions" rule is required — this was a reasonable default in the earlier sketch but is correctly reclassified here as one option among several, to be justified empirically rather than hard-coded. That the representation used for discovery must be identical to whatever gets used in the eventual propensity model.

Validation/checkpoint: The role-candidacy report exists and is carried forward unchanged in content (only refined, not overwritten) through Phases 7, 9, and 10 — `z_ecg`'s final role should be traceable to specific evidence at each stage, not asserted at the end.

Phase 6 — Background Knowledge and Causal Constraints

Objective: Encode what is *already known with justification* as hard constraints on the discovery algorithms, and be explicit about what is deliberately left open for the data to inform — per the critique, intuitive plausibility alone is not sufficient justification for a forced edge.

Inputs: Phases 1–5 outputs.

What we do — stated explicitly, not left implicit:

- **Forbidden edges (structural, not merely plausible):** outcome → any other variable (outcome is terminal, by construction of the estimand in Phase 0). Any variable classified "ineligible/post-time-zero" in Phase 3 → any baseline-eligible variable (temporal impossibility, not a judgment call).
- **Required tier ordering (temporal, justified by Phase 1/2, not intuition):** {demographics, comorbidities, baseline-eligible `_before` treatments} precede {the treatment variable(s) fixed in Phase 0} precede {index-ECG features and `z_ecg`, *only if* Phase 5 confirmed pre-treatment timing — otherwise this tier placement is itself an open question, not a constraint} precede {outcome}.
- **Left open for discovery (explicitly, with reasoning for why no constraint is justified):** the existence/direction of edges among comorbidities, demographics, and ECG features where no temporal or structural fact forces an ordering — e.g., whether hypertension and ECG morphology share an edge, and in which direction, is not something Phase 1–5 can settle, and is exactly what Phase 7 is for.
- Document every constraint with its justification (temporal fact vs. structural necessity vs. "left open") in a single table, so that nothing is forced into the graph "because it seems right."

Decision/output produced: A constraint specification file (forbidden edges, required tiers, explicitly-open relationships) to be passed directly into the discovery algorithms.

What must NOT be assumed: That plausible-sounding directions (e.g., "baseline ECG physiology → treatment") should be hard-constrained without a specific temporal/structural justification — per the critique, this should be left to discovery + literature + expert review unless Phase 1–3 genuinely forces it.

Validation/checkpoint: Every constraint in the specification traces to a specific Phase 1–5 finding, not to intuition.

Phase 7 — Causal Discovery (Stage A: Algorithmic Discovery)

Objective: Generate *candidate* graph structures consistent with the data and the Phase 6 constraints. This stage produces hypotheses, not conclusions — restated deliberately, because this is the critique's central correction to the earlier sketch.

Inputs: Phase 4 dataset, Phase 6 constraints.

What we do, and why, per method:

- **PC (constraint-based):** Infers structure via conditional-independence testing. Assumes causal sufficiency (no unmeasured confounders) and faithfulness. You have known candidate unmeasured confounders (smoking, lipid panel, BMI — none present in the dataset), so causal sufficiency is **known to be violated** in the full clinical sense; PC's output must be interpreted under that caveat, not treated as if the assumption holds.
- **FCI:** Relaxes causal sufficiency, explicitly allowing for latent confounders; output is a PAG (partial ancestral graph) with some edges left as genuinely ambiguous (circle marks) rather than forced into a direction. Given the known-missing-confounder situation, FCI's output is arguably the more honest primary result, even though it's less immediately usable than a fully-directed DAG.
- **GES (score-based):** A useful cross-check against the constraint-based methods, since it can be more

stable with moderate sample sizes and makes a different set of assumptions (score equivalence class search) — agreement between GES and PC/FCI, or lack of it, is informative precisely because the methods fail in different ways.

- **LiNGAM / NOTEARS (supplementary, optional):** Only run if there's a specific reason to want fully-directed output and a willingness to document the additional assumptions (linearity + non-Gaussianity for LiNGAM; the optimization formulation's own assumptions for NOTEARS). These remain supplementary evidence, not part of the primary pipeline, consistent with the critique.

What each method can and cannot identify: PC/GES assume no unmeasured confounding and can mis-orient or spuriously connect edges when that assumption fails (your situation). FCI can represent the presence of latent confounding explicitly but cannot fully resolve it — some edges will remain genuinely undetermined, and that undetermined status should be preserved into Phase 9/10 rather than resolved by picking a direction. None of these methods can establish actionability, clinical plausibility, or whether an edge reflects a real mechanism versus an artifact of this specific cohort's selection.

Handling disagreement — per the critique, explicitly not majority vote: when PC, GES, and FCI disagree on an edge's existence or direction, this is diagnostic information pointing to one or more of: insufficient sample size for that specific relationship, a genuinely latent confounder affecting that edge, representation issues (especially plausible for any edge touching `z_ecg`), or temporal-structure misspecification carried in from Phase 1 constraints. Document each disagreement with a hypothesis about *which* of these it most likely reflects, to be checked in Phases 8–10, rather than resolved by picking whichever algorithm "wins."

Decision/output produced: Three (or four, if LiNGAM/NOTEARS run) separate candidate graphs, plus a disagreement log with hypothesized causes per disputed edge. No graph is yet treated as more authoritative than another.

What must NOT be assumed: That any single algorithm's output is "the" discovered DAG. That disagreement should be averaged/voted away. That causal sufficiency holds for PC/GES just because the algorithm runs without error.

Validation/checkpoint: A disagreement log exists and every disputed edge has at least a working hypothesis before proceeding to Phase 8.

Phase 8 — Stability and Sensitivity Analysis (Stage B: Stable Candidate Structure)

Objective: Determine which edges are statistically stable under resampling — a necessary filter before spending literature/expert effort on an edge, but explicitly **not** a causality criterion, per the critique.

Inputs: Phase 7 outputs, Phase 4 dataset.

What we do:

- Bootstrap at the **unit level matching Phase 1's causal unit** (resample patients, not rows, if any residual row-level correlation structure exists) and re-run each Phase 7 algorithm across resamples.
- Treat the **number of bootstrap iterations as a configurable, justified parameter** rather than a fixed 100–200 default: start with a moderate number, check whether edge-frequency estimates have stabilized (e.g., frequency estimates stop materially changing with more iterations), and increase if not, bounded by computational cost — document the chosen number and the stability check that justified it.
- Compute edge frequency per candidate edge, per algorithm

- Compute edge frequency per candidate edge, per algorithm.
- **State explicitly what an edge-frequency threshold does and does not mean:** a high bootstrap frequency indicates the edge is a *statistically stable feature of this dataset under this method's assumptions* — it says nothing about whether the assumptions themselves hold (causal sufficiency, faithfulness, no unmeasured confounding for that specific edge), and therefore nothing conclusive about causal truth. A frequency threshold is used here only to **prioritize which edges receive literature/expert scrutiny**, not to certify them.
- Additionally run basic sensitivity checks on the Phase 5 representation choice for `z_ecg` if a reduction was used: does the stable-edge set change materially under an alternative reduction? This is itself part of this phase's sensitivity analysis, not a separate step.

Decision/output produced: A stable-candidate graph per algorithm (edges above the chosen frequency threshold), with frequencies reported numerically (not just a pass/fail cut), and a short note on `z_ecg`-representation sensitivity.

What must NOT be assumed: That "stable under bootstrap" means "causally confirmed" — explicitly rejected here, as the critique required. That a single fixed threshold (e.g., 70–80%) is universally correct — report frequencies and justify whatever threshold is used for prioritization in this specific analysis.

Validation/checkpoint: Every edge carried into Phase 9 has a documented frequency and algorithm-agreement status, not just a binary "included."

Phase 9 — Literature / Physiology Audit (Stage C: Scientific Interpretation)

Objective: Assess plausibility of the statistically stable candidate edges against established knowledge — and, per the critique's refinement, recognize that literature already did some work earlier (Phase 6's constraints) and now does a second, different job here: evaluating specific discovered/stable edges rather than imposing structure wholesale.

Inputs: Phase 8 stable-candidate graphs, Phase 6 constraint rationale (for consistency-checking), domain literature resource (e.g., via NotebookLM over the available MI/ECG physiology material).

What we do:

- For every edge that survived Phase 8's stability filter, assess it against the literature and categorize it: **data+literature supported**, **data-supported/literature unclear**, **literature-supported/data-unclear** (i.e., physiologically expected but not statistically stable here — informative about possible power limitations), **contested** (literature and data disagree on direction or existence), or **rejected** (literature strongly contradicts the edge, e.g., an implausible direction that likely reflects an artifact).
- Pay specific, separate attention to every edge touching `z_ecg` — given Phase 5's explicit caution, these edges need the clearest documented reasoning for their category, since "the literature doesn't discuss an uninterpretable 64-dim embedding" is a common and expected outcome that should be logged as **data-supported/literature silent**, not forced into either supported or rejected.
- Note explicitly: a literature source (including an LLM-assisted review of a specific textbook) reflects the scope and any limitations of that source material — it is a tool for efficient review, not an independent validator, and does not substitute for the clinical expert's judgment about this specific cohort and dataset in Phase 10.

Decision/output produced: A literature-audited graph: every Phase 8 stable edge tagged with one of the five categories above, plus a written note for every edge touching `z_ecg` specifically.

What must NOT be assumed: That literature silence equals rejection, or that literature support equals proof for an edge the data doesn't support (the critique correctly warns against "forcing the DAG to reproduce the literature" — a literature-supported-but-data-unclear edge should be flagged as a gap for the expert to weigh in on, not inserted into the graph by literature fiat).

Validation/checkpoint: No edge reaches Phase 10 without an explicit category and justification.

Phase 10 — Expert Validation (Stage D)

Objective: Formal clinical review of the literature-audited graph — treated here as a structured, documented process, not an informal final check, per the critique.

Inputs: Phase 9 output, Phase 0–5 specification documents (estimand, causal unit, time zero, treatment/outcome definitions, `z_ecg` role-candidacy report).

What the expert receives:

- The full specification stack from Phases 0–5 (not just the graph in isolation).
- The literature-audited graph with every edge's category and the per-category reasoning.
- A specific, written question set (below).

What the expert must answer (structured, recorded verbatim):

1. Is the causal unit (patient/admission) and time-zero definition clinically sound for this question?
2. Is the treatment definition (including the `_before` eligibility verdicts from Phase 2) clinically meaningful, or does it conflate a real decision with a severity marker?
3. Is the outcome definition appropriate, given its documented provenance (or its still-open status)?
4. Is the Phase 1 temporal ordering correct, in the expert's clinical experience with this kind of data?
5. For each **contested** or **data-only** edge: does clinical knowledge support, contradict, or remain genuinely uncertain about it?
6. Specifically for `z_ecg`: given the role-candidacy report and its empirical association findings (Phase 5), does the expert consider a confounder/proxy role plausible, or is an outcome-severity-only / predictive-only role more consistent with their understanding?
7. Is the resulting adjustment-set candidate (to be finalized in Phase 11) clinically defensible, or does it omit something the expert considers essential (even if unmeasured in this dataset) or include something inappropriate?

What happens when the expert disagrees with the statistical/literature result:

- Document the disagreement explicitly: which edge, what the data/literature said, what the expert said, and the expert's stated reasoning.
- Default resolution principle: **expert clinical judgment overrides statistical/literature evidence for questions of clinical mechanism and actionability** (e.g., "is this a real intervention," "is this variable a severity marker") — because these are not questions the data or literature alone can answer for this

specific cohort. Statistical/literature evidence should override intuition-only expert disagreement only where the expert's objection is itself empirically checkable and checked (rare, but should be allowed for in principle).

- Any such override, in either direction, is logged with its reasoning — not silently resolved.

Decision/output produced: An expert-validation record (structured answers + disagreement log) and a revised graph incorporating the resolutions.

What must NOT be assumed: That the expert's role is a rubber stamp at the end — this phase can send work back to earlier phases (e.g., a disagreement about treatment eligibility sends you back to Phase 2, not just to a DAG edit).

Validation/checkpoint: Every disagreement has a documented resolution and reasoning before Phase 11 begins.

Phase 11 — Final DAG and Adjustment Set (Stage E: Final Causal DAG)

Objective: Produce the single graph that will actually be used for identification — explicitly distinguished from every prior intermediate graph, per the critique's point that raw algorithmic / stable / literature-audited / expert-reviewed graphs are not interchangeable and only the last should drive adjustment.

Inputs: Phase 10 output.

What we do:

- Finalize the DAG, incorporating all Phase 10 resolutions.
- Apply the backdoor criterion to identify all backdoor paths from the Phase 0 treatment to the Phase 0 outcome.
- Derive the minimal sufficient adjustment set; explicitly list excluded variables and why (post-treatment/descendant, collider, or genuinely irrelevant per the final DAG).
- Resolve `z_ecg`'s final adjustment-set membership specifically here, citing the Phase 5/9/10 evidence chain that supports whatever role it was ultimately given (included as confounder-candidate, included as effect-modifier-only, or excluded from adjustment but retained for prediction/other use).

Decision/output produced: The final DAG (versioned, distinct from every intermediate graph, each of which is retained for audit trail, not discarded) and the adjustment-set specification.

What must NOT be assumed: That the final DAG is "done" in the sense of being beyond revision if Phase 12 diagnostics reveal a problem (e.g., severe positivity violation might require revisiting whether a variable truly belongs in the adjustment set, or whether the treatment/population definition needs narrowing).

Validation/checkpoint: The adjustment set traces fully to the final DAG via an explicit backdoor-path listing, not to a general "these seem like the right confounders" judgment.

Phase 12 — Pre-Estimation Causal Diagnostics

Objective: Check the identification assumptions the final DAG and adjustment set imply, *before* any effect estimation — per the critique's point 10, ATE/CATE should not start until these are checked, not just until the

DAG exists.

Inputs: Phase 11 output, Phase 4-equivalent full dataset (not the discovery-reduced representation necessarily — this may use the full `z_ecg` dimensionality if that's what Phase 11 assigned to the adjustment set).

What we do:

- Fit a preliminary propensity model on the finalized adjustment set; check overlap/positivity (propensity distribution by treatment group, extreme-score prevalence).
- Check covariate balance diagnostics conceptually ahead of the full treatment-model build (Phase 13), to catch gross problems early.
- Reassess consistency (well-defined treatment versions) and temporal-ordering assumptions one final time against the finalized definitions.
- Explicitly document known threats to exchangeability (unmeasured confounders absent from the dataset, e.g., smoking/lipids/BMI) as a standing limitation, not something this phase can resolve.

Decision/output produced: A diagnostics report: positivity verdict (adequate / needs trimming / fails for this population), consistency/temporal-ordering confirmation, and a documented unmeasured-confounding limitations statement.

What must NOT be assumed: That passing these diagnostics proves no unmeasured confounding — it cannot; it only confirms the checkable assumptions (positivity, measured balance) are not obviously violated.

Validation/checkpoint: Positivity is adequate (post any necessary trimming, with the trimmed population explicitly described) before Phase 13 begins.

Phase 13 — ATE/CATE Estimation

Objective: Only now does effect estimation begin, using the finalized adjustment set and diagnostics-cleared treatment model — this phase is intentionally described only briefly here, since it was covered in detail in the prior conversation (propensity model → ATE via doubly robust/DML methods → CATE via causal forest/DML-family methods with the Phase 5/11-resolved `z_ecg` role determining whether it enters as a control (`W`) or an effect-modifier (`X`) argument → counterfactual estimates framed under the Phase 12-documented assumptions).

What must NOT be assumed: That this phase can begin if any earlier phase's checkpoint failed or was left unresolved.

Final Methodological Judgment

1. Is this overall strategy scientifically defensible? Yes, with the restructuring above. The original discovery → literature → expert sketch was directionally reasonable but under-specified two things the critique correctly identified: (a) data-semantics decisions (unit, time zero, treatment/outcome eligibility) were at risk of being entangled with or run after causal discovery rather than strictly before it, and (b) the intermediate graphs (raw algorithmic, stable, literature-audited, expert-reviewed) were at risk of being collapsed into one "the DAG," losing the evidentiary trail that makes the final adjustment set defensible. With Phases 0–3 strictly preceding

Phase 4's dataset construction, and Phases 7–11 kept as distinct, non-collapsing stages, the strategy is methodologically sound for a DAG-and-adjustment-set deliverable feeding into ATE/CATE.

2. Which parts of the original roadmap should be retained? The core method choice (PC + GES as primary, FCI for the known-latent-confounding situation, LiNGAM/NOTEARS as optional supplements), the general idea of bootstrap stability analysis, and the sequencing principle of discovery-before-literature-before-expert for the *interpretation* of specific edges.

3. Which parts should be changed?

- `z_ecg` dimensionality reduction: changed from a near-default instruction ("reduce to 5–10 dimensions") to an empirically-justified, explicitly optional choice with mandatory sensitivity reporting if used (Phase 5).
- Bootstrap edge-frequency interpretation: changed from an implicit "high frequency = causal" framing to an explicit statement that frequency indicates statistical stability under stated assumptions only, used solely to prioritize edges for Phase 9/10 scrutiny (Phase 8).
- Algorithm disagreement handling: changed from (implicitly) something to be smoothed over to an explicit diagnostic trigger with a documented hypothesis per disputed edge (Phase 7).
- Literature's role: changed from a single post-discovery step to two distinct jobs — informing hard constraints before discovery (Phase 6) and assessing specific stable edges after (Phase 9) — without letting either job override the other's evidence by default.
- Expert validation: changed from an end-stage review into a formally structured stage with a fixed question set, an explicit disagreement-resolution principle, and the possibility of sending work back to earlier phases (Phase 10).
- Graph versioning: changed from implicit to explicit — raw, stable, literature-audited, expert-reviewed, and final DAG are now five distinct, separately retained artifacts (Phase 7–11), per the critique's point 9.

4. Which parts should be removed? The instruction to treat causal-discovery output as directly informing basic data-semantics questions (unit, time zero, treatment eligibility) — these are now strictly Phases 1–3, upstream of and inaccessible to the discovery algorithms. The implicit suggestion that a single DAG-building pass could serve multiple candidate causal questions interchangeably (Phase 0 now requires this to be fixed, or repeated per candidate).

5. What are the biggest remaining methodological risks?

- **Known unmeasured confounders** (smoking, lipid panel, BMI, and possibly others not in the column list) that no amount of discovery/literature/expert work inside this dataset can fix — this is a standing limitation to carry into the final report, not a solvable step.
- `z_ecg` **'s causal role may remain genuinely ambiguous** even after Phases 5/9/10 — a 64-dimensional learned representation may simply resist clean causal categorization, and the project should be prepared to report this ambiguity honestly (e.g., "included as a covariate for balance improvement without a confirmed confounder justification, with that distinction stated explicitly") rather than forcing a clean answer.
- **Treatment-eligibility verification (Phase 2) may be limited by what timestamp evidence actually exists** in the source data — if `_before` truly cannot be verified beyond naming convention, this caps how confidently any downstream causal claim can be made, regardless of how rigorous Phases 4–13 are.
- **Outcome-definition resolution is an external dependency** (Phase 0/2) that this roadmap cannot itself

resolve — it depends on clarifying the labeling-strategy history with whoever defined `Label_Troponin` / `is_MI` /the dataset ground truth.

6. What decisions must be made before any causal discovery algorithm is run? Causal unit (Phase 1); time zero and index-ECG rule (Phase 1); treatment eligibility verdicts (Phase 2); outcome definition or an explicit, bounded placeholder (Phase 2); baseline-covariate eligibility (Phase 3); the discovery-specific `z_ecg` representation choice with its justification (Phase 5); and the background-knowledge constraint specification (Phase 6). None of these are the discovery algorithm's job, and none should be inferred from its output.

7. What should be considered exploratory versus confirmatory? Exploratory: the raw algorithmic graphs (Phase 7), any edge categorized data-only/literature-unclear or contested (Phase 9) that the expert does not resolve with high confidence (Phase 10), and any CATE heterogeneity discovered outside the pre-specified effect-modifier list in Phase 13. **Confirmatory:** the final expert-validated DAG and adjustment set (Phase 11) used for the primary ATE estimate, and any CATE result along a pre-specified, expert-endorsed effect modifier. These two categories should be labeled as such in every subsequent report and paper draft — conflating them is one of the more common ways causal-discovery-assisted analyses overstate their conclusions.